USER-DISSATISFACTION-BASED STATIC BIKE REPOSITIONING: FORMULATIONS, SOLUTION METHODS, AND EXTENSIONS TO BROKEN BIKES AND ON-SITE REPAIR

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Ph.D. Thesis

The University of Hong Kong

June, 2026

Abstract of thesis entitled

"User-Dissatisfaction-Based Static Bike Repositioning: Formulations, Solution Methods, and Extensions to Broken Bikes and On-Site Repair"

Submitted by

Runqiu HU

for the degree of Doctor of Philosophy

at The University of Hong Kong

in June, 2026

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User-Dissatisfaction-Based Static Bike Repositioning: Formulations, Solution Methods, and Extensions to Broken Bikes and On-Site Repair

by

Runqiu Hu

B.Eng.; M.Eng.

A Thesis Submitted in Partial Fulfilment
of the Requirements for the Degree of
Doctor of Philosophy

at

The University of Hong Kong.
June, 2026

Declaration

I, Runqiu Hu, declare that this thesis titled, "User-Dissatisfaction-Based Static Bike Repositioning: Formulations, Solution Methods, and Extensions to Broken Bikes and On-Site Repair", which is submitted in fulfillment of the requirements for the Degree of Doctor of Philosophy, represents my own work except where due acknowledgement have been made. I further declared that it has not been previously included in a thesis, dissertation, or report submitted to this University or to any other institution for a degree, diploma or other qualifications.

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Date: June 17, 2026

Hu Runqiu

Acknowledgements

I would like to thank...

Runqiu Hu
University of Hong Kong
June 17, 2026

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List of Abbreviations

ASAP	**A**s **S**oon **A**s **P**ossible
AKA	**A**lso **K**nown **A**s
BPFA	**B**eta **P**rocess **F**actor **A**nalysis
CNN	**C**onvolutional **N**eural **N**etwork
GAN	**G**enerative **A**dversarial **N**etwork
MSE	**M**ean **S**quare **E**rror
PSNR	**P**eak **S**ignal-to-**N**oise **R**atio
SSIM	**S**tructural **SIM**ilarity

List of Symbols

Global notations

I^{SR}	super-resolved light field image	—
I^{LR}	low-resolution light field image	—
I^{HR}	high-resolution light field image	—
E^{SR}	super-resolved epipolar plane image	—
E^{HR}	high-resolution epipolar plane image	—

Chapter 1

θ, ϕ	incoming direction expressed in term of spherical coordinates	rad
τ	time	s (second)
x,y	spatial coordinates with two-plane parameterization	1 (uint)
s,t	angular coordinates with two-plane parameterization	1 (uint)
P	radiance distribution	$\mathrm{W/srm^2Hz}$
Ω	image plane	—
Θ	parameters of the multi-layer framework	—
γ_s, γ_a	scaling factors of spatial / angular coordinates	1 (uint)
$\mathcal{L}$	loss function	—

Chapter 2

F_0	shallow features extracted by a single HConv layer	—
$F_{\mathrm{G_d}}$	feature maps extracted by the d^{th} HRB of the GRLNet	—
$H^{\mathrm{n}}_{\mathrm{HRB}}$	the operation of the n^{th} HRB of the SReNet	—
H_{AGBN}	the operation of the proposed aperture group batch normalization algorithm	—
H_{up}	upsampling operation on the low-resolution features	—
ℓ_A	angular loss	—
ℓ_S	spatial perceptual loss	—

ℓ_{SA}	the weighted combination of ℓ_A and ℓ_S	—
f	the summation of all the feature maps after every activation function of VGG network	—
g	learned mapping between the low-resolution and high-resolution light field images	—

Chapter 3

x,y	spatial coordinates with two-plane parameterization	1 (uint)
s,t	angular coordinates with two-plane parameterization	1 (uint)
γ_s, γ_a	scaling factors of spatial / angular coordinates	1 (uint)
ℓ_G	generator adversarial loss	—
ϕ	denotes the mapping of VGG network	—
δ	nearest neighbor downsampling operator	—
κ	a Gaussian blurring kernel with a window size of 7×7 and standard deviation of 1.2 pixels	—
η	additive noise with zero mean and unit standard deviation	—

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INTRODUCTION

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APPENDIX A

About Appendix

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APPENDIX B

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APPENDIX C

Appendix Title Here

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Bibliography

[1] A. Gershun. “The light field”. In: *Journal of Mathematics and Physics* 18.1–4 (1939), pp. 51–151. doi: 10.1002/sapm193918151.